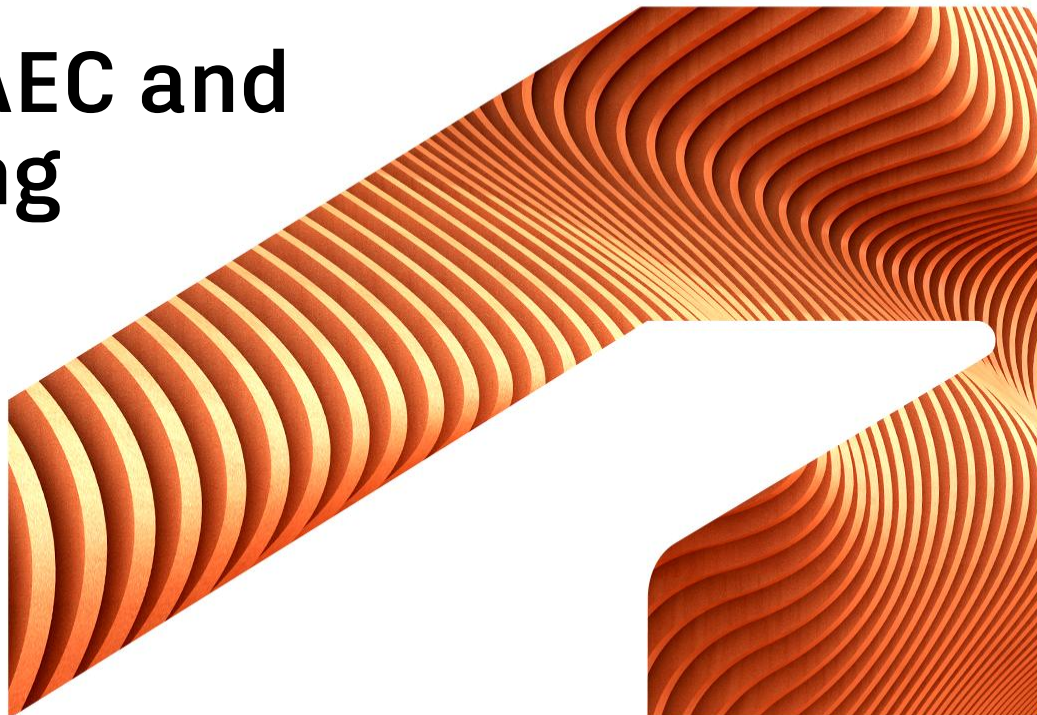


Spit it out: How AEC and the Manufacturing Data Model API can feed ERP Systems

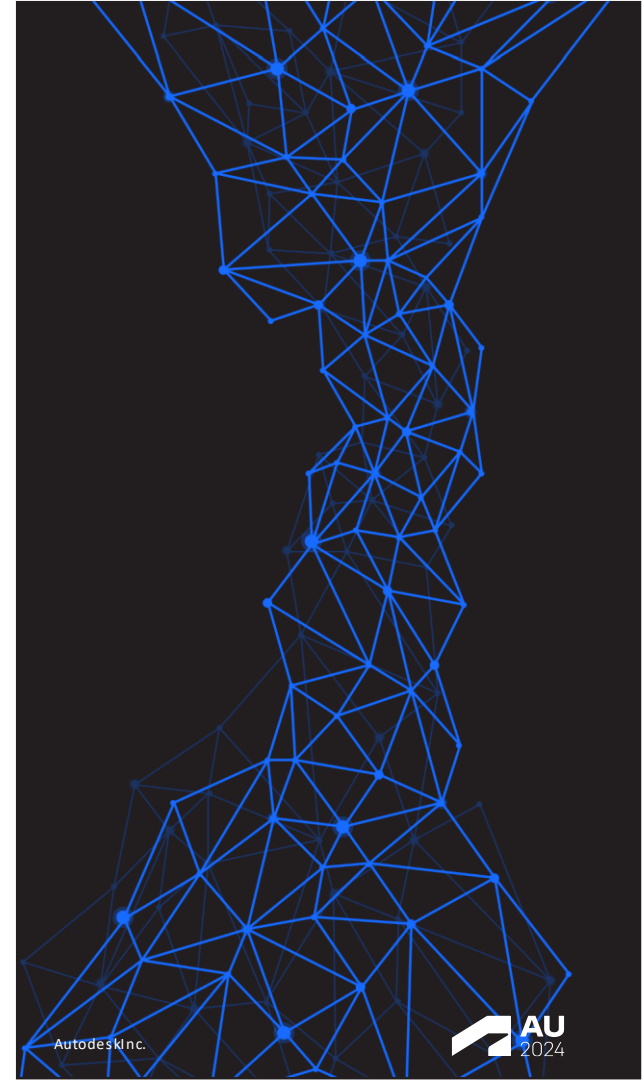
SD2336

Christian Gessner
Head of
Research & Innovation | @COOLORANGE



Agenda

- 1 Introduction
- 2 Autodesk Data Model & Granular Data
- 3 Manufacturing Data Model Project
- 4 AEC Data Model Project
- 5 Key Learnings & Limitations





COOLORANGE

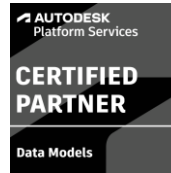
POWER UP YOUR DATA

Introduction

Who is COOLORANGE?

About COOLORANGE

- Founded in 2009
- By former Autodesk employees
- 25+ years in software development
- Our locations include Italy (HQ), Germany, Spain, Australia and USA
- 100% commitment to Autodesk



Experienced in Integrations



Projects in the Spotlight



 Fusion Add-In

 Items & BOM Transfer

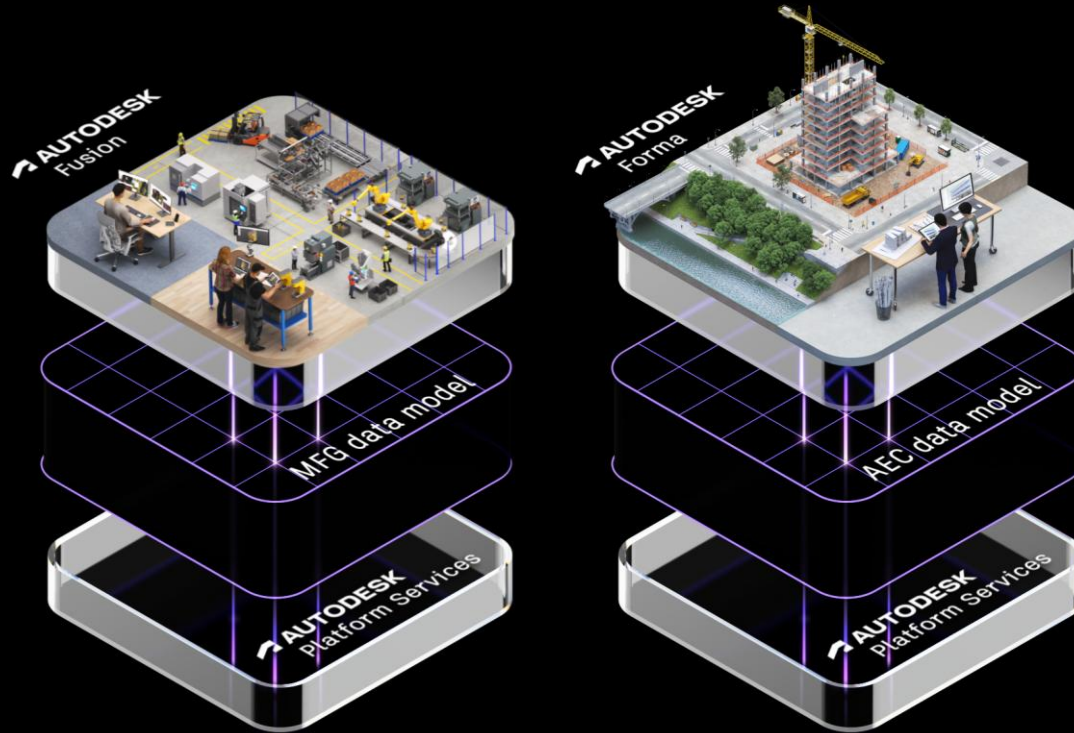


 Windows Task Scheduler

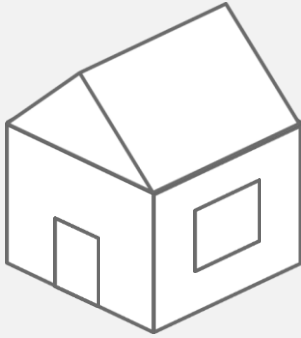
 ERP Delivery Creation

AUTODESK
Platform Services

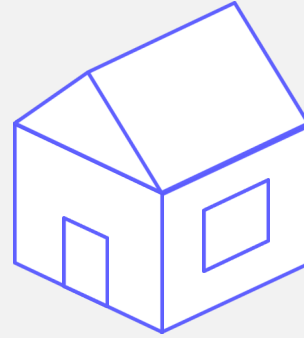
Autodesk Data Model – A Fileless Future



Files vs. Autodesk Data Model



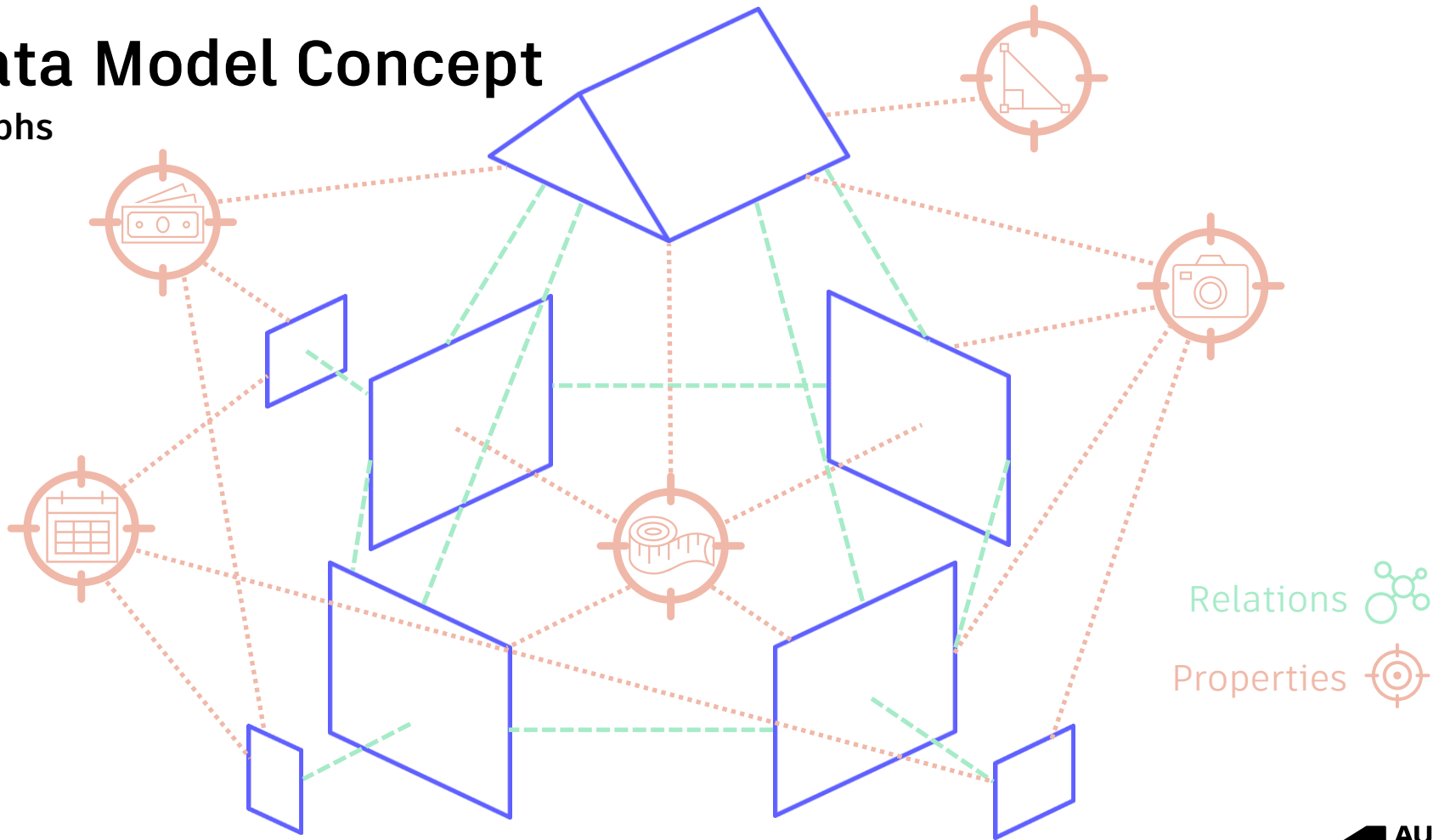
Building as a File



Building as a Graph

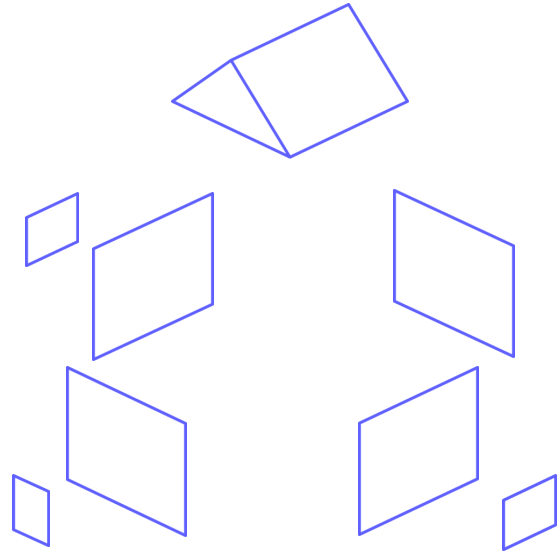
Data Model Concept

Graphs



Data Model Queries

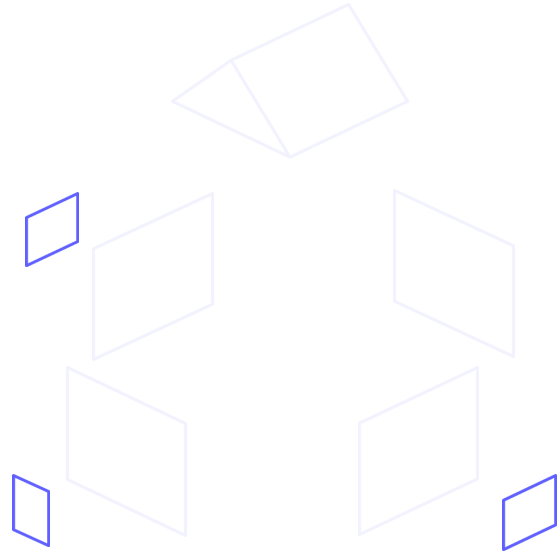
GraphQL



Data Model Queries

GraphQL

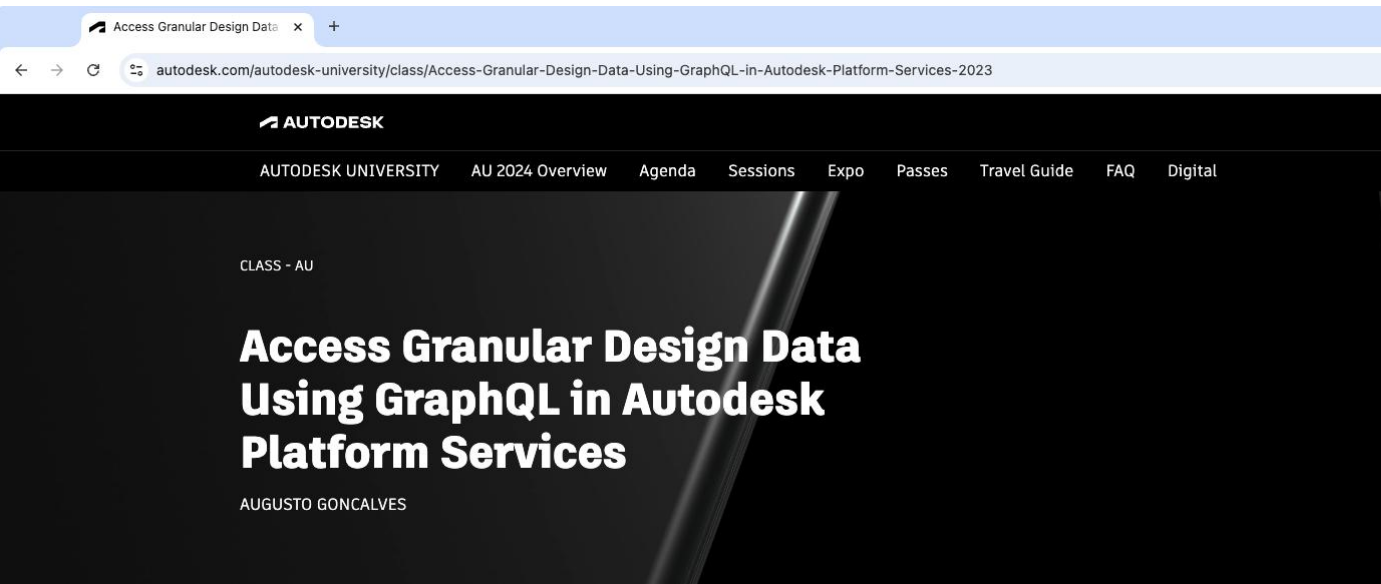
```
...  
query GetAllWindows {  
  elements(filter: {query:"type==window"}) {  
    results {  
      name  
      properties {  
        width  
        height  
        fire-rating  
      }  
    }  
  }  
}
```



Autodesk Data Model in Detail

AU Online

- Access Granular Design Data Using GraphQL in Autodesk Platform Services by Augusto Goncalves (APS Developer Relations & Advocacy)
- Explains the Data Model in all details!



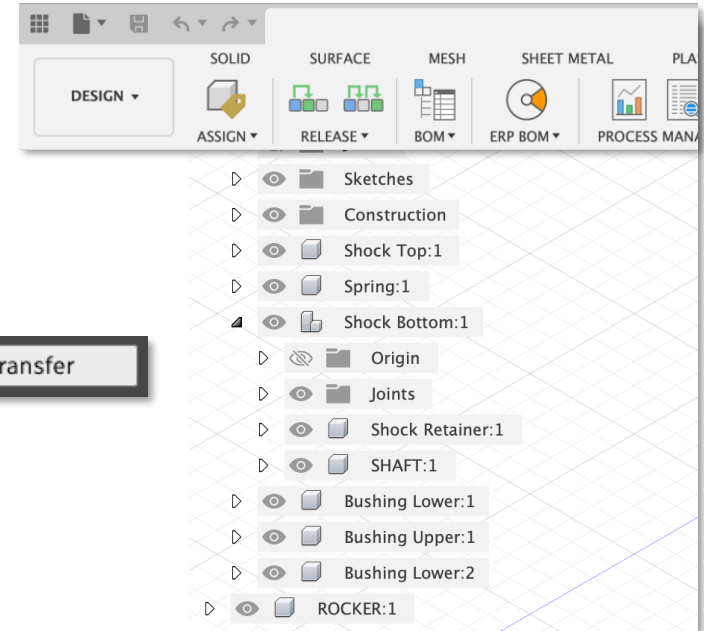


MFG Data Model

Fusion - Microsoft Business Central
Integration

Customer Requirements

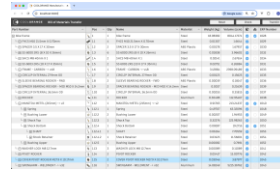
- Target system: Microsoft Business Central
- User Interface in Fusion
- Data
 - Component Structure & Quantities
 - Properties, Physical Properties & Thumbnails
- On-Demand
 - Compare Items and BOMs
 - Created / Updated Items and BOMs in ERP
- Part Numbers created in ERP



Solution Overview (APIs)



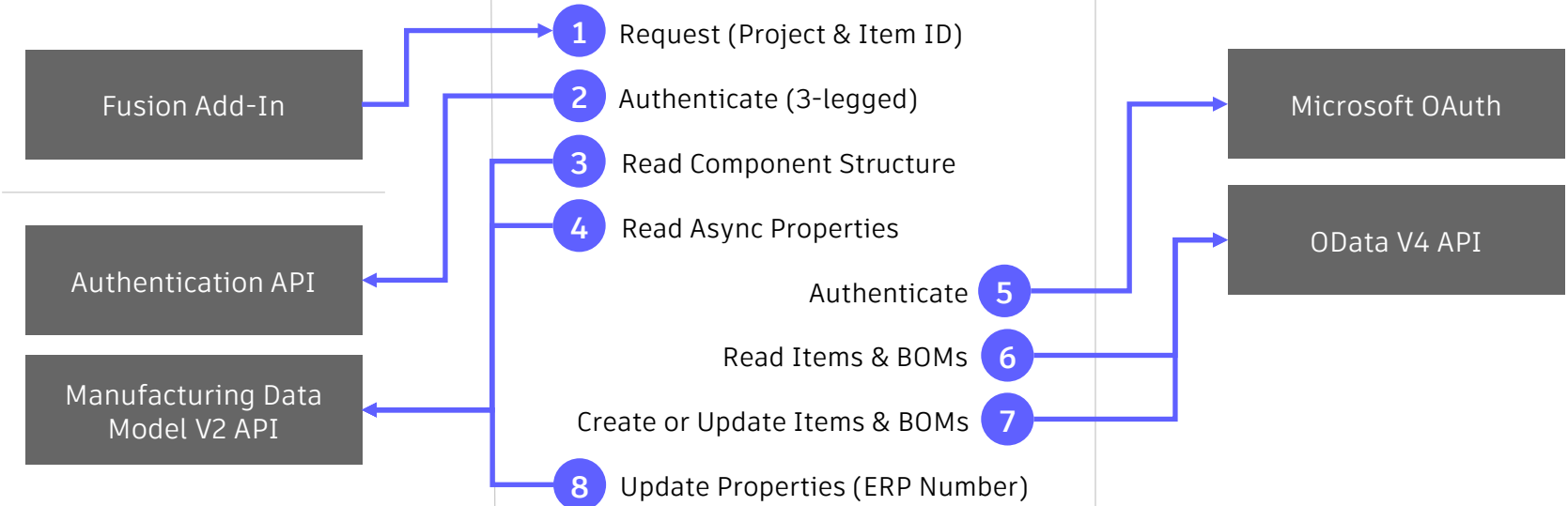
Fusion / Fusion Teams



C# / JS Web App



Business Central



Demo

No. ↑	Description	Type	Quantity on Hand	Substi... Exist	Assem... BOM	Production BOM No.	Routing No.	Base Unit of Measure	Cost is Adj...	Unit Cost	Unit Price
1896-S	ATHENS Desk	Inventory	4	No	No			PCS	✓	780.70	1,000.80
1900-S	PARIS Guest Chair, black	Inventory	0	No	No			PCS	✓	150.30	192.80
1906-S	ATHENS Mobile Pedestal	Inventory	5	No	No			PCS	✓	338.20	433.60
1908-S	LONDON Swivel Chair, blue	Inventory	3	No	No			PCS	✓	148.10	190.10
1920-S	ANTWERP Conference Table	Inventory	10	No	No			PCS	✓	505.40	647.80
1925-W	Conference Bundle 1-6	Inventory	0	No	Yes			PCS	✓	0.00	188.80
1928-S	AMSTERDAM Lamp	Inventory	8	No	No			PCS	✓	42.80	54.90
1929-W	Conference Bundle 1-8	Inventory	0	No	Yes			PCS	✓	0.00	233.80
1936-S	BERLIN Guest Chair, yellow	Inventory	100	No	No			PCS	✓	150.30	192.80
1953-W	Guest Section 1	Inventory	-49	No	Yes			PCS	✓	0.00	125.80
1960-S	ROME Guest Chair, green	Inventory	2	No	No			PCS	✓	150.30	192.80
1964-S	TOKYO Guest Chair, blue	Inventory	4	No	No			PCS	✓	150.30	192.80
1965-W	Conference Bundle 2-8	Inventory	-81	No	Yes			PCS	✓	0.00	233.80
1968-S	MEXICO Swivel Chair, black	Inventory	10	Yes	No			PCS	✓	148.10	190.10
1969-W	Conference Package 1	Inventory	-7	No	Yes			PCS	✓	0.00	341.80
1972-S	MUNICH Swivel Chair, yellow	Inventory	0	No	No			PCS	✓	148.10	190.10
1980-S	MOSCOW Swivel Chair, red	Inventory	0	Yes	No			PCS	✓	148.10	190.10
1988-S	SEOUL Guest Chair, red	Inventory	0	Yes	No			PCS	✓	150.30	192.80
1996-S	ATLANTA Whiteboard, base	Inventory	10	No	No			PCS	✓	1,089.90	1,397.30
2000-S	SYDNEY Swivel Chair, green	Inventory	38	No	No			PCS	✓	148.10	190.10
F-100	Remote pump	Inventory	0	No	No			PCS	✓	65.00	100.00

Details

📎 Attachments (0)

Item Details - Invoicing

Item No.	1896-S
Costing Method	FIFO
Cost is Adjusted	Yes
Cost is Posted to G/L	Yes
Standard Cost	780.70
Unit Cost	780.70
Overhead Rate	0.00
Indirect Cost %	0
Last Direct Cost	780.70
Profit %	21.99241
Unit Price	1,000.80

Item Details - Planning

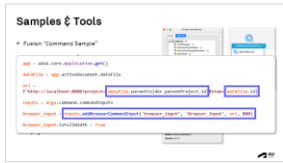
Item No.	1896-S
Reordering Policy	
Reorder Point	25
Reorder Quantity	0
Maximum Inventory	0
Overflow Level	0
Time Bucket	
Lot Accumulation Period	

Project Approach

Autodesk APIs

ERP APIs

Occurrences



Samples & Tutorials

Fusion “Command Sample”
APS “Hubs Browser”
MFG Data Model Samples
MFG Data Model Explorer
Voyager



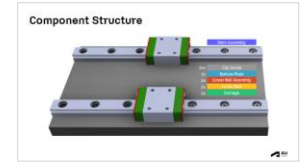
MFG Data Model GraphQL

Nested Queries
Pagination / Cursor
Fragments
Aliases, Unions
Mutations
Queries in C#



OAuth V4

Authentication (OAuth)
Metadata
Deep & Granular Data
Updates

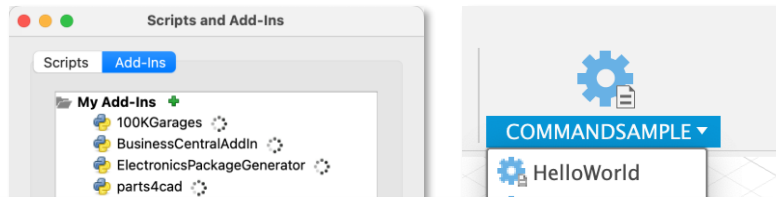


Component Structure

Tree Structure
Positions
Quantities

Samples & Tools

- Fusion “Command Sample”



```
app = adsk.core.Application.get()

dataFile = app.activeDocument.dataFile

url =
f'http://localhost:8080?project={dataFile.parentFolder.parentProject.id}&item={dataFile.id}'

inputs = args.command.commandInputs

browser_input = inputs.addBrowserCommandInput('browser_input', 'Browser Input', url, 800)

browser_input.isFullWidth = True
```

Author = Patrick Rainsberry
Version = 0.1.0

Samples & Tools

- Fusion “Command Sample”
- APS “Hubs Browser”
- Manufacturing Data Model Samples
 - Read the model hierarchy of a design
 - Get the thumbnail of a part
 - Retrieve Physical Properties of a Component
- MFG Data Model Explorer
- Voyager

The screenshot displays the MFG Data Model Explorer web application. The interface includes a sidebar on the left with a search bar and a list of entity types: Query, Application, BasicItem, BasicItemVersion, BasicItemVersions, BoundingBox3D, and Component. The main area shows a complex data model diagram with tables and their relationships. The tables shown are:

DesignItem	
id	ID!
hub	Hub
tpVersion	DesignItemVersion
thumbnail	Thumbnail
rootComponent	Component
tpRootComponentVersion	ComponentVersion
isConfiguration	Boolean
fusionWebUrl	Url
project	Project
parentFolder	Folder
name	String
createdOn	DateTime
createdBy	User
lastModifiedOn	DateTime
lastModifiedBy	User
versions	DesignItemVersions
extensionType	String

BasicItem	
id	ID!
hub	Hub
project	Project
parentFolder	Folder
name	String
createdOn	DateTime
createdBy	User
lastModifiedOn	DateTime
lastModifiedBy	User
tpVersion	BasicItemVersion
versions	BasicItemVersions
extensionType	String

The diagram shows relationships between these tables, with lines connecting fields in one table to fields in another. For example, the 'versions' field in DesignItem is connected to the 'tpVersion' field in BasicItem. The 'parentFolder' field in DesignItem is connected to the 'parentFolder' field in BasicItem. The 'project' field in DesignItem is connected to the 'project' field in BasicItem. The 'createdOn' field in DesignItem is connected to the 'createdOn' field in BasicItem. The 'createdBy' field in DesignItem is connected to the 'createdBy' field in BasicItem. The 'lastModifiedOn' field in DesignItem is connected to the 'lastModifiedOn' field in BasicItem. The 'lastModifiedBy' field in DesignItem is connected to the 'lastModifiedBy' field in BasicItem. The 'tpVersion' field in DesignItem is connected to the 'tpVersion' field in BasicItem. The 'versions' field in DesignItem is connected to the 'versions' field in BasicItem. The 'extensionType' field in DesignItem is connected to the 'extensionType' field in BasicItem.

At the bottom of the interface, there are checkboxes for 'Sort by Alphabet', 'Skip Relay', 'Skip deprecated', and 'Show leaf fields', along with a 'RESET' button.

GraphQL Queries

```
1 query GetTipRootComponentVersion($hubId: ID!, $itemId: ID!, $propName: [String!]!) {  
2   item(hubId: $hubId, itemId: $itemId) {  
3     ... on DesignItem {  
4       tipRootComponentVersion {  
5         ... commonProps  
6         allOccurrences pagination: {limit: 50} {  
7           pagination {  
8             pageSize  
9             cursor  
10          }  
11         results {  
12           id  
13           parentComponentVersion {  
14             id  
15           }  
16           childComponentVersion componentVersion {  
17             ... commonProps  
18           }  
19         }  
20       }  
21     }  
22   }  
23 }  
24 }  
25
```

```
26 fragment commonProps on ComponentVersion {  
27   id  
28   lastModifiedOn  
29   name  
30   partNumber  
31   partDescription  
32   materialName  
33   customProperties(filter: {names: $propName})  
34   results {  
35     name  
36     value  
37   }  
38 }  
39 }
```



```
1 query GetTipRootComponentVersion($hubId: ID!, $itemId: ID!, $propName: [String!]!) {  
2   item(hubId: $hubId, itemId: $itemId) {  
3     ... on DesignItem {  
4       tipRootComponentVersion {  
5         ... commonProps  
6         allOccurrences(pagination: {limit: 50}) {  
7           pagination {  
8             pageSize  
9             cursor  
10          }  
11          results {  
12            id  
13            parentComponentVersion {  
14              id  
15            }  
16            childComponentVersion: componentVersion {  
17              ... commonProps  
18            }  
19          }  
20        }  
21      }  
22    }  
23  }  
24 }
```

Variables Headers

```
1 {  
2   "hubId": "urn:adsk.ace:prod.scope:4848fe8d-73f2-4ead-b104-c9f0d6c58963",  
3   "itemId": "urn:adsk.wipprod.dm.lineage:XXysPoGRIquUjcwYF50aw",  
4   "propName": "ERP Number"  
5 }
```

GraphQL Queries

```
1 query GetTipRootComponentVersion($hubId: ID! $itemId: ID!, $propName: [String!]!) {
2   item(hubId: $hubId, itemId: $itemId) {
3     ... on DesignItem {
4       tipRootComponentVersion {
5         ... commonProps
6         allOccurrences(pagination: {limit: 50}) {
7           pagination {
8             pageSize
9             cursor
10          }
11          results {
12            id
13            parentComponentVersion {
14              id
15            }
16            childComponentVersion: componentVersion {
17              ... commonProps
18            }
19          }
20        }
21      }
22    }
23  }
24 }
25
```

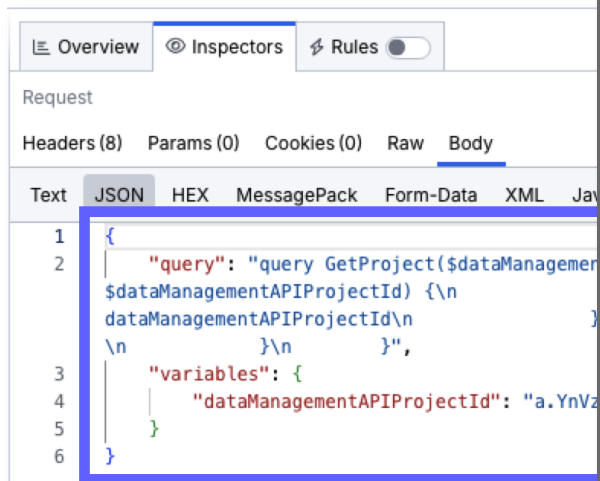
```
1 query GetProjectByDataManagementAPIId($projectId: ID!) {
2   projectByDataManagementAPIId(dataManagementAPIProjectId: $projectId) {
3     hub {
4       id
5       name
6       propertyDefinitionCollections {
7         results {
8           name
9           definitions {
10            results {
11              name
12              id
13            }
14          }
15        }
16      }
17    }
18  }
19 }
```

Variables	Headers
1 {	
2 "input": {	
3 "targetId": "Y29tcH5EWmJ5ZnFYRDJlR1lQMw5WYTZhZ1A4X0wyQ"	
4 "propertyInputs": {	
5 "propertyDefinitionId": "cHJvcGRlZn5WMVpaR3M2MUFBa2E"	
6 "value": "Number from the ERP system"	
7 }	
8 }	
9 }	

GraphQL Queries (C#)

Payload

```
{  
  "query": "<GRAPHQL QUERY>  
  "variables": <VARIABLES>  
}
```



```
7 references  
public async Task<T> Query<T>(string query, dynamic variables, string bearerToken)  
{  
    var request = new RestRequest(_endpoint, Method.Post); //"beta/graphql"  
  
    request.AddHeader("Accept", "application/json");  
    request.AddHeader("Authorization", $"Bearer {bearerToken}");  
  
    query = query.Replace(Environment.NewLine, "\\n");  
    query = query.Replace("\\\"", "\\\"");  
  
    var variableJson = JsonConvert.SerializeObject(variables, Formatting.Indented);  
    var json = $"{{\"query\": \"{{query}}\", \"variables\": {{variableJson}} }}";  
  
    request.AddParameter("application/json", json, ParameterType.RequestBody);  
    var response = await _client.ExecuteAsync<T>(request);  
    if (response.IsSuccessStatusCode)  
    {  
        if (response != null && response.Data != null)  
            return response.Data;  
    }  
  
    return default;  
}
```

Authentication (OAuth 2.0)

Overview Authorization • Scripts Variables Runs

This authorization method will be used for every request in this collection. You can override this by specifying one in the request.

Configure New Token

Token Name

Business Central Token

Grant type

Client Credentials

Access Token URL ⓘ

https://login.microsoftonline.com/2f8...

https://login.microsoftonline.com/{{TENANT_ID}}/oauth2/v2.0/token

Client ID ⓘ

00a1f193-c533-4fa5-9c5e-9ed7f...

Client ID

Client Secret ⓘ

jnf8Q-83erWTG2qKLLcWfu-V_Ai...

Client Secret

Scope ⓘ

https://api.businesscentral.dynamics...

https://api.businesscentral.dynamics.com/.default

Client Authentication ⓘ

Send client credentials in body

OData - Metadata

GET

https://api.businesscentral.dynamics.com/v2.0/00000000-0000-0000-0000-000000000000/AU2024/ODATAV4/\$metadata#Company('CRONUS%20USA%2C%20Inc.')

Send

ParamsAuthorizationHeaders (9)BodyScriptsSettings

BodyCookiesHeaders (17)Test Results

200 OK • 556 ms • 55.55 KB • Save Response

PrettyRawPreviewVisualizeXML

```
5500 <EntityType Name="ProductionBOMs">
5501 <Key>
5502 <PropertyRef Name="No" />
5503
5504
5505 <Property Name="No" Type="Edm.String" Nullable="false" MaxLength="20">
5506 <Annotation Term="NAV.LabelId" String="No." />
5507 <Annotation Term="NAV.NavType">
5508 | <EnumMember>NAV.NavType/String</EnumMember>
5509 </Annotation>
5510 <Annotation Term="NAV.AllowEditOnCreate" Bool="true" />
5511 </Property>
5512 <Property Name="Description" Type="Edm.String" MaxLength="100">--
5513 </Property>
5514 <Property Name="Unit_of_Measure_Code" Type="Edm.String" Nullable="false" MaxLength="10">
5515 </Property>
5516 <Property Name="Status" Type="Edm.String">--
5517 </Property>
5518 <Property Name="Search_Name" Type="Edm.String" MaxLength="100">--
5519 </Property>
5520 <Property Name="Version_Nos" Type="Edm.String" MaxLength="20">--
5521 </Property>
5522 <Property Name="ActiveVersionCode" Type="Edm.String">--
5523 </Property>
5524 <Property Name="Last_Date_Modified" Type="Edm.Date">--
5525
5526 <NavigationProperty Name="ProductionBOMsProdBOMLine" Type="Collection(NAV.ProductionBOMsProdBOMLine)" Partner="ProductionBOMs" ContainsTarget="true">
5527 <ReferentialConstraint Property="No" ReferencedProperty="Production_BOM_No" />
5528 </NavigationProperty>
5529 <Annotation Term="NAV.LabelId" String="ProductionBOMs" />
5530 </EntityType>
```

OData - Deep & Granular Data

GET `https://api.businesscentral.dynamics.com/v2.0/{{TenantId}}/{{Environment}}/ODataV4/Company('{{Company}}')/ProductionBOMs('1001')?$expand=ProductionBOMsProdBOMLine($select=Production_BOM_No,Version_Code,Line_No,No,Position,Description,Quantity_per)&$select=No,Description,Unit_of_Measure_Code`

Params • Authorization Headers (7) Body Scripts Settings

Query Params

☒	Key	Value
☒	\$expand	ProductionBOMsProdBOMLine(\$select=Production_BOM_No,Version_Code,Line_No,No,Position,Description,Quantity_per)
☒	\$select	No,Description,Unit_of_Measure_Code

```
1
2 "@odata.context": "https://api.businesscentral.dynamics.com/v2.0/{{TenantId}}/{{Environment}}/ODataV4/$metadata#Company('CRONUS%20USA%2C%20Inc.')
```

```
3 "@odata.etag": "W/\\"JzE50zMzNzYwNzA1MjY5NjU4NDANzZkx0ZAw0yc=\"",
4 "No": "1001",
5 "Description": "bike frame",
6 "Unit_of_Measure_Code": "PCS",
7 "ProductionBOMsProdBOMLine": [
8   {
9     "@odata.etag": "W/\\"JzIw0zEwMDg5MzY5NzUxMTc5NzE4MjI5MTswMDsn\"",
10    "Production_BOM_No": "1001",
11    "Version_Code": "",
12    "Line_No": 10000,
13    "No": "1003",
14    "Description": "FHCS M10 21.5mm X 0.70mm",
15    "Quantity_per": 4,
16    "Position": "1"
17  },
18  {
19    "@odata.etag": "W/\\"JzIw0zE30Tc0Mzk00DA40DI4MTQ2NzgwMTswMDsn\"",
20    "Production_BOM_No": "1001",
21    "Version_Code": ""
```

OData - Updates

PATCH ▼ `https://api.businesscentral.dynamics.com/v2.0/{{TenantId}}/{{Environment}}/ODataV4/Company('{{Company}}')/ItemCards('1165')`

Params Authorization **Headers (10)** Body ● Scripts Settings

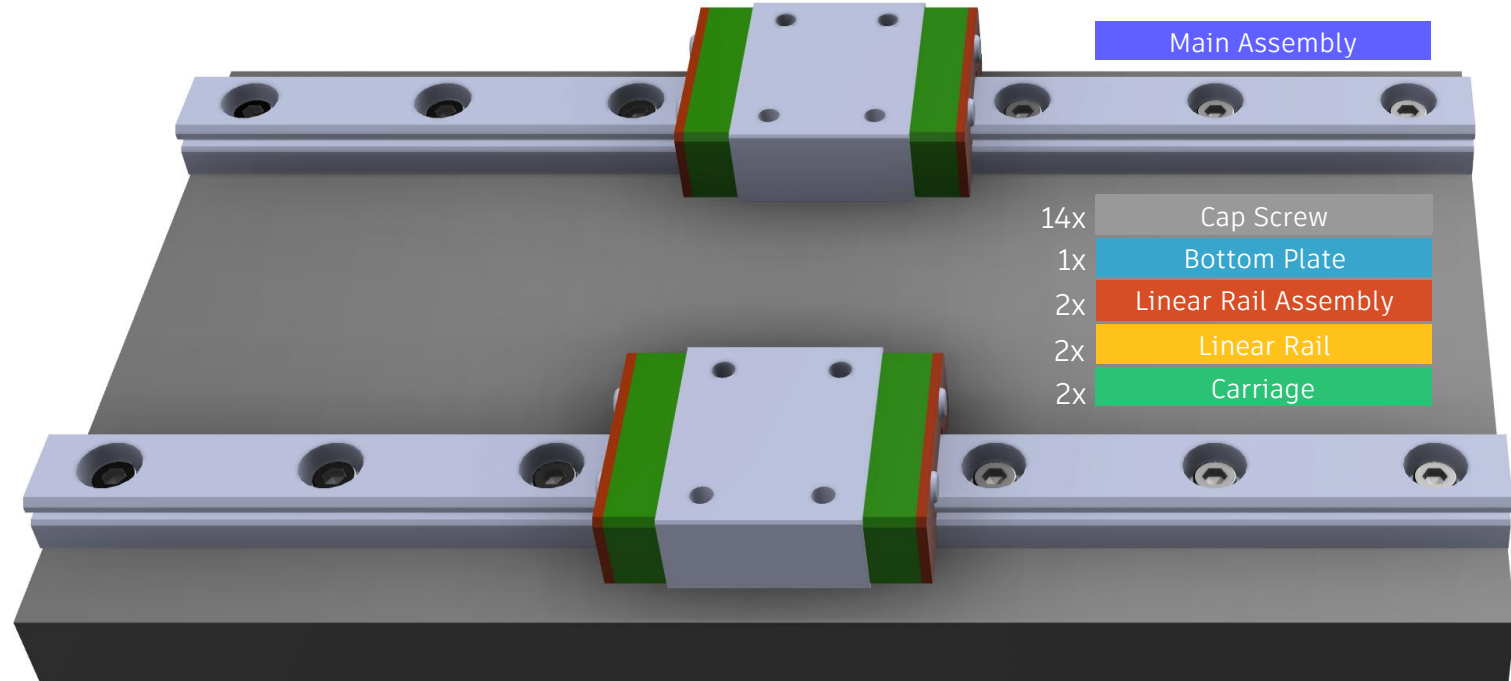
Key	Value
☒ Accept-Encoding	gzip, deflate, br
☒ Connection	keep-alive
☒ If-Match	W/"JzIwOzEzNjUxNjM3ODM5NjgwMjk3NjAwMTswMDsn"

Body Cookies Headers (18) Test Results 201 Created

Pretty Raw Preview Visualize JSON ▼ ↔

```
1 {
2   "@odata.etag": "W/\"JzIwOzEzNjUxNjM3ODM5NjgwMjk3NjAwMTswMDsn\"",
3   "Description": "TEST ITEM created with Postman",
4   "Description_2": "",
5   "Blocked": false,
6   "Type": "Inventory",
7   "Base_Unit_of_Measure": "PCS",
8 }
9
```

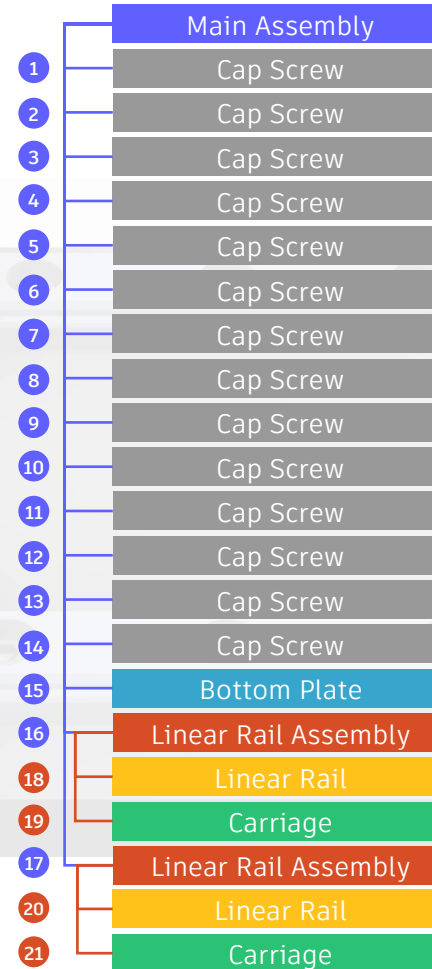
Component Structure



Component Structure

Occurrence	
id	ID!
isDistributedDesign	Boolean
componentVersion	ComponentVersion
parentComponentVersion	ComponentVersion
occurrences	Occurrences!

Id	Parent Component Version	Component Version
1	Main Assembly	Cap Screw (1)
2	Main Assembly	Cap Screw (2)
...	...	...
14	Main Assembly	Cap Screw (14)
15	Main Assembly	Bottom Plate
16	Main Assembly	Linear Rail Assembly (1)
17	Main Assembly	Linear Rail Assembly (2)
18	Linear Rail Assembly (1)	Linear Rail (1)
19	Linear Rail Assembly (1)	Carriage (1)
20	Linear Rail Assembly (2)	Linear Rail (2)
21	Linear Rail Assembly (2)	Carriage (2)



Component Structure

- Tree structure from flat GraphQL results
- Quantities
- Positions

1				Main Assembly	
1	.	1		Cap Screw	14x
1	.	2		Bottom Plate	1x
1	.	3		Linear Rail Assembly	2x
1	.	3	.	Linear Rail	1x
1	.	3	.	Carriage	1x

COOLORANGE - BILL OF MATERIALS TRANSFER					
COOLORANGE Bill of Materials Transfer					
Part Number	Pos	Qty	Name	Material	
[-] Main Assembly	1	1	Main Assembly	Steel	
[-] [-] ANSI B18.3.1M - M1.6x0.35...	1.1	14	Broached Hexagon Socket Head Cap Screw ANSI B18.3.1M - M1.6x0.35 x 8 Steel Grade 2 Plain	Steel Gr	
[-] [-] Bottom Plate	1.2	1	Bottom Plate	Steel	
[-] [-] Linear Rail Assembly	1.3	2	LinearRail-LWL7B-Assy	Steel	
[-] [-] [-] Carriage-LWL7B	1.3.1	1	Carriage LWL7B	Steel	
[-] [-] [-] LinearRail-LWL7B-100...	1.3.2	1	LinearRail LWL7B 100mm	Steel	



AEC Data Model

ACC/Revit – Odoo Inventory
Automated Deliveries

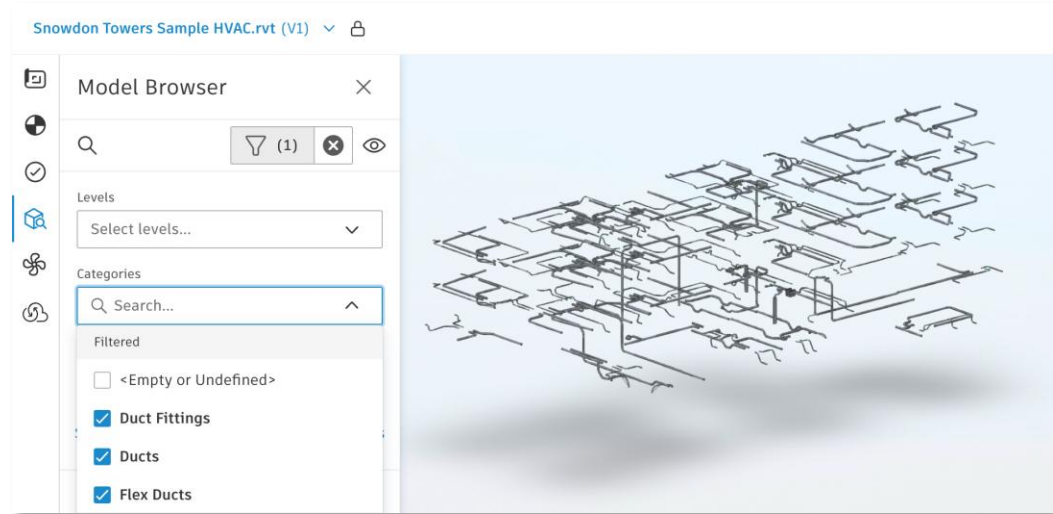
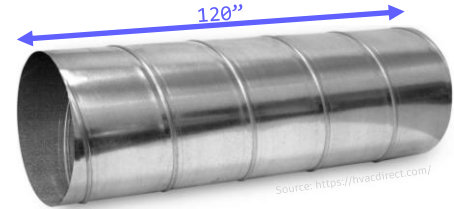
Customer Requirements

- Target system: Odoo (Inventory)
- ACC Submittals
 - References Revit Designs
 - Trigger: Closed / Approved
 - “Lock” Revit Files
- Source Data
 - Ducts and Fittings
 - Pipes in 120” length
- Target Data
 - Products
 - Deliveries & Product Quantities

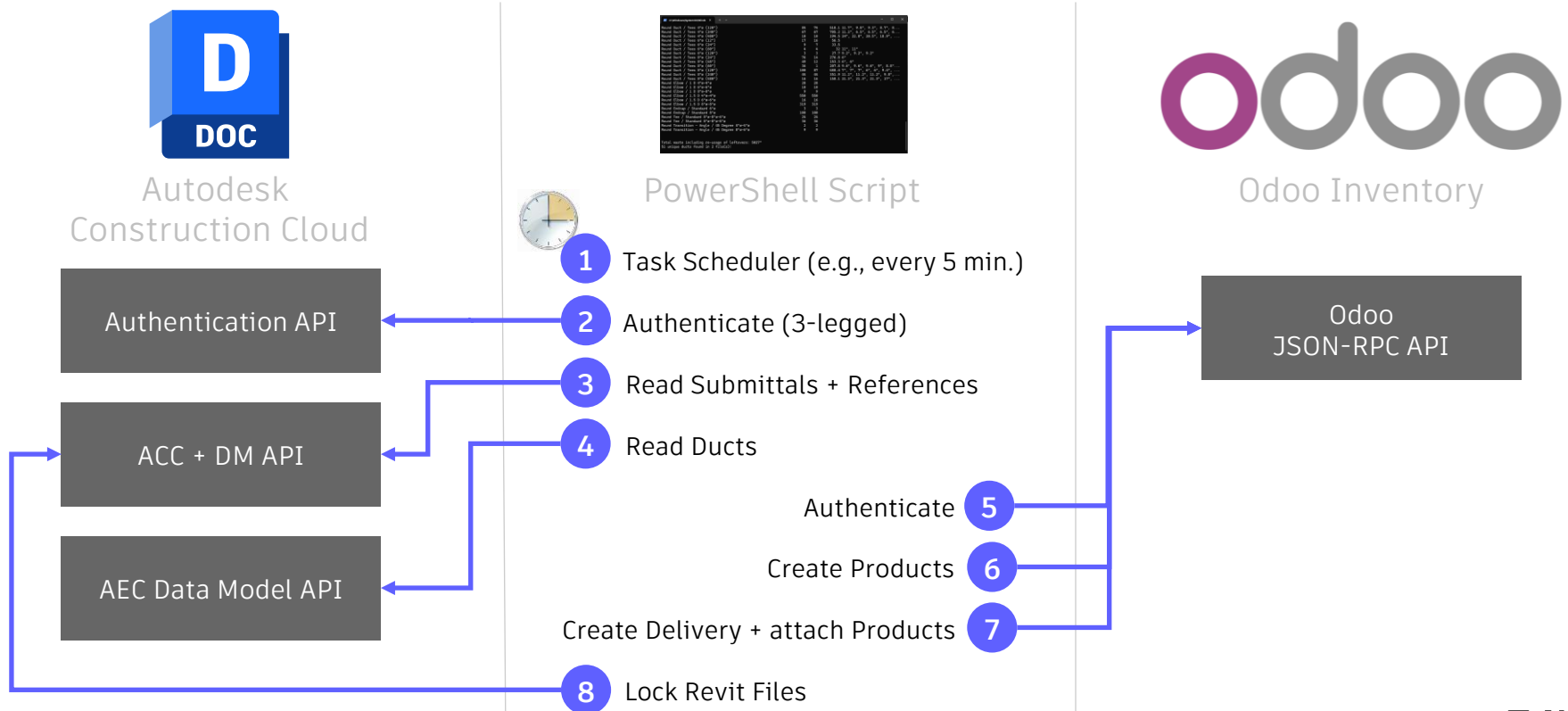
odoo

Submittal

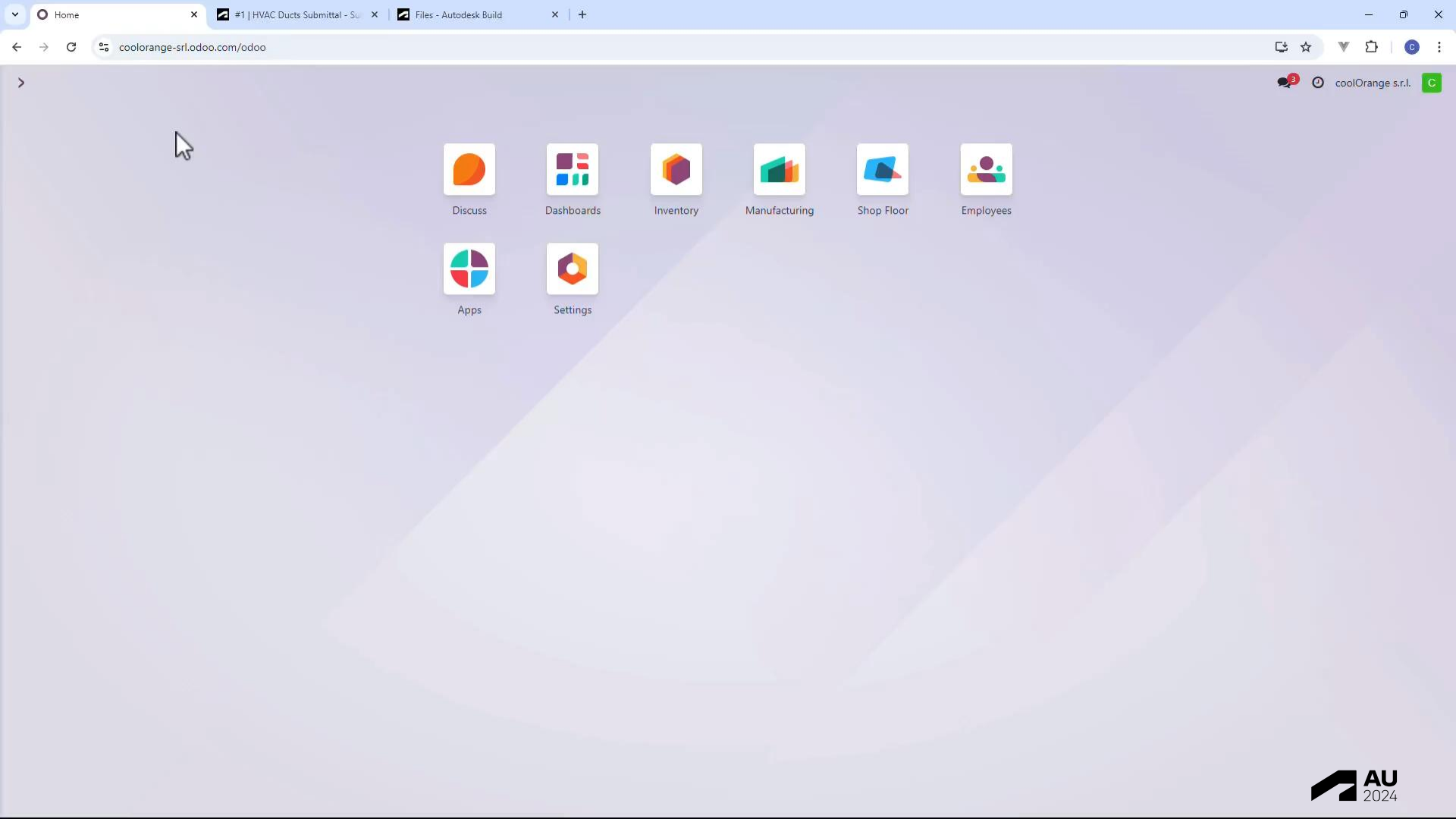
Closed Approved



Solution Overview (APIs)



Demo



Project Approach

Autodesk APIs

ERP APIs

User-less



Samples & Tools

Quantity Takeoff Tutorial
AEC Data Model Explorer
Voyager



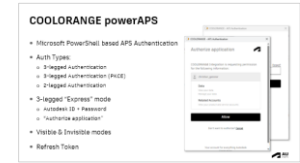
AEC Data Model GraphQL

Nested Queries
Pagination / Cursor
Query Filter
Properties Filter
Queries in PowerShell



JSON-RPC

Authentication (JSON-RPC)
API Calls
Invoke-WebRequest
Invoke-RestMethod



APS Authentication

COOLORANGE
powerAPS



Samples & Tools

- AEC Data Model API Tutorial
 - Quantity Takeoff Example
- AEC Data Model Explorer
- Voyager

The screenshot displays the Autodesk Platform Services AEC Data Model Explorer web application. The interface is divided into a left sidebar and a main content area. The sidebar, titled 'Type List', contains a search bar and a list of schema types with their descriptions:

- Query**: The entry-point for Autodesk entity queries. This acts as the top-level API from which all queries must start.
- DistinctPropertyValue**: A PropertyValue and its count.
- DistinctPropertyValues**: Contains a list of DistinctPropertyValue returned in response to a query.
- DistinctPropertyValuesCollection**: A collection of distinct properties matching the name given.
- Element**: Represents an element type.
- ElementAlternativeIdentifiers**: Contains alternative identifiers for an element.
- ElementGroup**: Represents a Revit model.
- ElementGroupAlternativeIdentifiers**: Contains alternative identifiers for an AEC elementGroup.
- ElementGroupExtractionStatus**: Information about elementGroup extraction status.
- ElementGroupVersion**: Represents a single version of an ElementGroup.
- ElementGroupVersionHistory**: Information related to versions of an elementGroup.
- ElementGroupVersions**

The main content area features a complex graph visualization showing the relationships between these entities. At the bottom of the main area, there is a 'Query' input field and a set of checkboxes for query options: 'Sort by Alphabet' (unchecked), 'Skip Relay' (checked), 'Skip deprecated' (checked), and 'Show leaf fields' (checked). A 'RESET' button is located at the bottom right of the interface.

GraphQL Queries

```
1 query GetElementGroup($projectId: ID!, $fileUrn: [String!]) {
2   elementGroupsByProject(projectId: $projectId, filter: {fileUrn: $fileUrn}) {
3     results {
4       name
5       id
6     }
7   }
8 }

1 query GetProject($hubId: ID! $projectId: ID! $fileUrn: [String!]) {
2   projects(hubId: $hubId, filter: {projectId: $projectId, fileUrn: $fileUrn}) {
3     results {
4       name
5       id
6     }
7   }
8 }

1 query GetHub($hubName: String!) {
2   hubs(filter: {name: $hubName}) {
3     results {
4       name
5       id
6     }
7   }
8 }

1 query GetElementGroup($hubName: String!, $projectName: String!, $fileUrn: [String!]) {
2   hubs(filter: {name: $hubName}, pagination: {limit: 1}) {
3     results {
4       name
5       id
6     }
7     projects(filter: {name: $projectName}, pagination: {limit: 1}) {
8       results {
9         name
10        id
11        elementGroups(filter: {fileUrn: $fileUrn}, pagination: {limit: 1}) {
12          results {
13            name
14            id
15          }
16        }
17      }
18    }
19  }
20 }
```



GetElementGroup x GetDuctsFromElementGroup +

Graph/QL

```
1 query GetElementGroup($hubName: String!, $projectName: String!, $fileUrn: [String!]!) {
2   hubs(filter: {name: $hubName}, pagination: {limit: 1}) {
3     results {
4       name
5       id
6       projects(filter: {name: $projectName}, pagination: {limit: 1}) {
7         results {
8           name
9           id
10          elementGroups(filter: {fileUrn: $fileUrn}, pagination: {limit: 1}) {
11            results {
12              name
13              id
14            }
15          }
16        }
17      }
18    }
19  }
20 }
```



Variables Headers

```
1 {
2   "hubName": "build_coolorange",
3   "projectName": "Aerospace Components Facility",
4   "fileUrn": "urn:adsk.wipprod:dm.lineage:alQ7KQhXS8yVN6bPw511jg"
5 }
```

GraphQL Queries (PowerShell)

6 references

```
function Invoke-GraphQLQuery($query, $variables) {  
    $body = @"  
{  
    "query": "$($query.Replace("`r`n", "\n"))",  
    "variables": $variables  
}  
"@  
    $parameters = @{  
        "Uri" = "https://developer.api.autodesk.com/aec/graphql"  
        "Method" = "Post"  
        "ContentType" = "application/json; charset=utf-8"  
        "Body" = $body  
        "Headers" = $ApsConnection.RequestHeaders  
    }  
    $parameters["Headers"].Add("Accept", "application/json; charset=utf-8")  
    $response = Invoke-RestMethod @parameters  
    $json = $response | ConvertTo-Json -Depth 10  
    $bytes = [System.Text.Encoding]::GetEncoding("ISO-8859-1").GetBytes($json)  
    $response = [System.Text.Encoding]::UTF8.GetString($bytes) | ConvertFrom-Json  
    return $response.data  
}
```

JSON-RPC API (Odooh)

Authentication

POST

https://{{ENDPOINT}}/web/session/authenticate

Send

ParamsAuthorizationHeaders (9)Body ●ScriptsSettingsCookies

none

form-data

x-www-form-urlencoded

raw

binary

GraphQL

JSON

Beautify

```
1 {
2   "jsonrpc": "2.0",
3   "method": "call",
4   "params": {
5     "db": "{{DATABASE}}",
6     "login": "{{USER}}",
7     "password": "{{PASSWORD}}",
8     "context": {}
9   },
10  "id": 1
11 }
```

BodyCookies (2)Headers (11)Test Results200 OK • 1109 ms • 2.18 KB • e.g. ⋮

Name	Value	Domain	Path	Expires	HttpOnly	Secure
frontend_lang	en_US	coolorange-s ...	/	Wed, 27 Aug ...	false	false
session_id	9cwGRtRIBAk...	coolorange-s ...	/	Mon, 23 Sep ...	true	true

JSON-RPC API (Odoo)

API Calls

POST ▼

Send ▼

Params Authorization Headers (9) Body ● Scripts Settings

Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON** ▼

Beautify

```
1  {
2    "jsonrpc": "2.0",
3    "method": "call",
4    "params": {
5      "model": "product.product",
6      "method": "search_read",
7      "args": [
8        [{"id", ">", 0}],
9        ["id", "name", "list_price", "qty_available", "uom_id", "uom_po_id", "categ_id"]
10     ],
11    "kwargs": {
12      "limit": 1000
13    }
14  },
15  "id": 2
16 }
```

Invoke-WebRequest (PowerShell)

PowerShell cmdlets

```
function Invoke-OdooAuthentication {  
    $body = @"  
    {  
        "jsonrpc": "2.0",  
        "method": "call",  
        "params": {  
            "db": "$odooDatabase",  
            "login": "$odooUsername",  
            "password": "$odooPassword",  
            "context": {}  
        },  
        "id": 1  
    }  
"@  
    $parameters = @{  
        "Uri" = "https://$odooDatabase.odoo.com/web/session/authenticate"  
        "Method" = "Post"  
        "ContentType" = "application/json; charset=utf-8"  
        "Body" = $body  
    }  
    $response = Invoke-WebRequest @parameters -SessionVariable odooSession  
    $json = $response.Content | ConvertFrom-Json
```

Invoke-RestMethod (PowerShell)

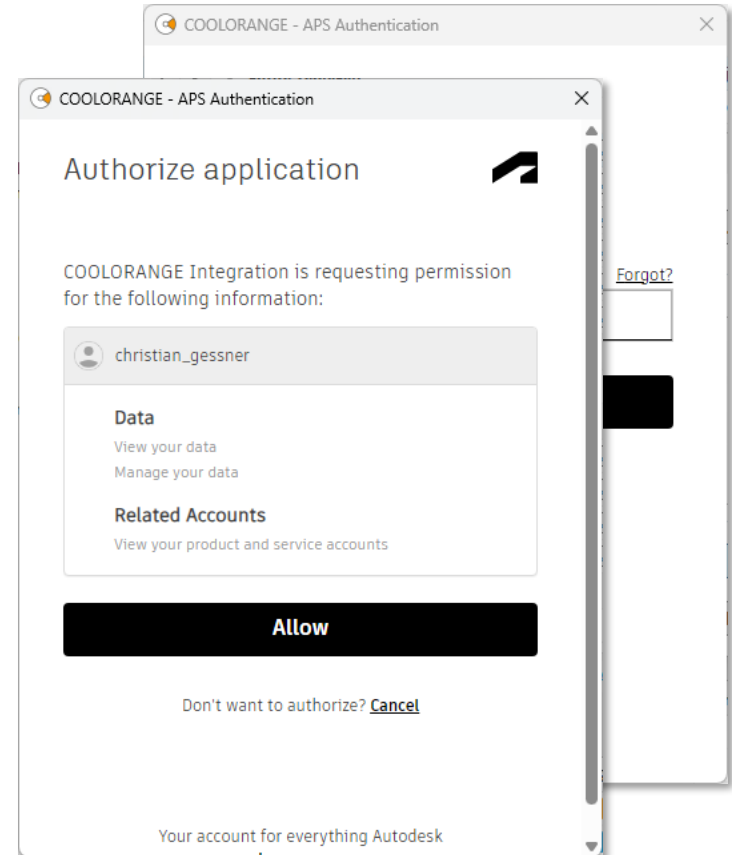
PowerShell cmdlets

```
function Invoke-OdooDataSetCall($body) {  
    $parameters = @{  
        "Uri" = "https://$odooDatabase.odoo.com/web/dataset/call_kw"  
        "Method" = "Post"  
        "ContentType" = "application/json; charset=utf-8"  
        "Body" = $body  
    }  
    $response = Invoke-RestMethod @parameters -WebSession $odooSession  
    return $response.result  
}
```

```
function Get-OdooProducts {  
    $body = @"  
    {  
        "jsonrpc": "2.0",  
        "method": "call",  
        "params": {  
            "model": "product.product",  
            "method": "search_read",  
            "args": {  
                [{"id", ">", 0}],  
                [{"id", "name", "list_price", "qty_available", "uom_id", "uon_po_id", "categ_id"}  
            ],  
            "kwargs": {  
                "limit": 1000  
            }  
        },  
        "id": 2  
    }  
    "@  
    return Invoke-OdooDataSetCall $body  
}
```


COOLORANGE powerAPS

- Microsoft PowerShell based APS Authentication
- Auth Types:
 - 3-legged Authentication
 - 3-legged Authentication (PKCE)
 - 2-legged Authentication
- 3-legged “Express” mode
 - Autodesk ID + Password
 - “Authorize application”
- Visible & Invisible modes
- Refresh Token



File

Edit

Selection

View

Go

Run

...

←

→

coolOrange

3-legged.ps1

3-legged PKCE.ps1

2-legged.ps1

Hubs.ps1

PowerShell Extension

Scripts > 3-legged.ps1 > ...

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

Import-Module powerAPS

3-Legged OAuth

\$parameters=@{

"ClientId"===="V1ZZGs61AAka5Beoaolu1NVbvXCGRHZU"

"ClientSecret"===="

"CallbackUrl"===="http://localhost:8080/api/auth/callback"

"Scope"===="data:read data:write account:read"

"Username"===="christian.gessner@coolorange.com"

"Password"===="

}

\$result=Open-APSCConnection @parameters -Express

if (-not \$result){

throw "Authentication failed"

}

\$ApsConnection

Close-APSCConnection

powerAPS: 3-legged...

Token

:

Bearer eyJhbGciOiJIUzI1NiIsImtpZCI6IHRUFpfSmoXz1TYzNZS01oRERBZFBWeFowOF9SUzI1NiIsInBpLmF0bSI6ImFzc2MiifQ.eyJzY29wZSI6WyJkYXRhOnJlYWQiLCJkYXRhOnR1IiwiaWF0IjoiYHR0cHM6Y9kZXZlbg69wZXIuYXBpLmF1dG9kZXNrLmNvbSI6ImF1ZCI6Imh0dHBzOi8vYXV0b2Rlc2suY29tIiwianRpIjoiwUFhaDdoeVJjdGdmU3F2Ym

Issued

:

9/8/2024 4:53:07 PM

Lifespan

:

4199

ExpiresIn

:

4199

ValidUntil

:

9/8/2024 6:03:06 PM

RefreshToken

:

8ifEubGOsbdUGsaP9PooNQX7mYIVRneIEKBmaKwmwr

ClientId

:

V1ZZGs61AAka5Beoaolu1NVbvXCGRHZU

AuthType

:

ThreeLegged

Scope

:

data:read data:write account:read

CallbackUrl

:

http://localhost:8080/api/auth/callback

Username

:

christian.gessner@coolorange.com

RequestHeaders

:

{Authorization}

powerAPS: Disconnect...

powerAPS: Revoke Token...

True

PS C:\ProgramData\coolOrange>

<

0

0

0

0

Ln 1, Col 1 (527 selected)

Spaces: 4

UTF-8

CRLF

{}

PowerShell

Prettier



Key Learnings & Limitations

Key Learnings & Limitations

Autodesk Data Model APIs

- Pagination and Cursor
- Standard Filtering
- Advanced Filtering using RSQL
- Rate Limits and Quotas
- Asynchronous Data
- Fragments
- Directives
- Aliases
- Mutations

APS Authentication

- Traditional Web App
 - 3-legged
- Desktop, Single-Page App
 - 3-legged with PKCE
 - No Client Secret needed
- Server-to-Server
 - 2-legged
 - "X-User-Id" header
 - Doesn't support most APIs
- Solution
 - COOLORANGE powerAPS
 - SSA (Secure Service Account) - (currently in Private Beta)

The screenshot displays the 'App Registrations' page in the Azure portal. Two applications are listed:

- aec/graphql**: Request ID 79d6... Authorization URI: http://localhost:8080/.oauth2/token
- beta/graphql**: Request ID 79d6... Authorization URI: http://localhost:8080/.oauth2/token; Redirect URI: http://localhost:8080/callback

Both applications have their 'Authorization URI' field highlighted with a red box.

Alternative Identifiers

Data Management IDs vs. Data Model IDs vs. ACC IDs

Object	Data Management API	Data Model APIs
Hub	b.561a49f-dbd9-4d8c-e8bf-4fac72b8d33a	urn:adm:ace-prod:scope:a3bcac42-drc2-42b5-Baa7-a3bcaebaf8fc
Project	b.835aa9f9-e8af-457f-b397-ea9dc0e44a42	urn:adm:acc-woriproc-prod:project:c013079-1a88-4ba3-b075-3c5fa27ef9af
File Version + Element Group	urn:adm:woriprod.fs.file.b5g1mleQcDz2Mzmsfw/version=2	WYV2w3vayM3ZG130a2R10dxVZ2M4SjHmM2gUxbyO3aaWtEci8W0L2VZ2yOdySyungscJ33

Object	Data Management API	Autodesk Construction Cloud API
Project	b.835aa9f9-e8af-457f-b397-ea9dc0e44a42	B355aa9f9-e8af-457f-b397-ea9dc0e44a42



Beta APIs

```
1  "errors": [
2
3    "message": "field 'itemId' argument 'projectId' of type 'ID!' is required, but it was not provided.",
4
5  ],
6
7  "data": {
8    "item": {
9      "itemId": "1",
10     "projectId": "1"
11   },
12   "itemNames": [
13     "name": "XXXXXXXXXXXXXXXXXXXX"
14   ],
15   "message": "Unknown argument 'projectId' as field 'itemNames'",
16
17 },
18
19 "errors": [
20
21   "message": "field 'itemId' argument 'projectId' of type 'ID!' is required, but it was not provided.",
22
23 ],
24
25 "data": {
26   "item": {
27     "itemId": "1",
28     "projectId": "1"
29   },
30   "itemNames": [
31     "name": "XXXXXXXXXXXXXXXXXXXX"
32   ]
33 }
```

Item(
 projectId: ID!
): Item

Item(
 hubid: ID!
 itemId: ID!
): Item

⚠ Beta APIs are subject to change. Please avoid using them in production environments.

AU
AUSTRALIAN
UNIVERSITY

APS Authentication

- Traditional Web App
 - 3-legged
- Desktop, Mobile, Single-Page App
 - 3-legged with PKCE
 - No Client Secret needed
- Server-to-Server
 - 2-legged
 - “X-User-Id” header
 - Doesn't support most APIs
- Solution
 - COOLORANGE powerAPS
 - SSA (Secure Service Account) - (currently in Private Beta)

POST aec/graphql

Request

Headers

Authorization *
string

Must be **Bearer** <TOKEN> , where <TOKEN> is an access token obtained by a [three-legged](#) OAuth flow.

POST beta/graphql

Request

Headers

Authorization *
string

Must be **Bearer** <TOKEN> , where <TOKEN> is an access token obtained by a [three-legged](#) OAuth flow.

GET projects/{projectId}/issues

Request

Headers

Authorization *
string

Must be **Bearer** <token> , where <token> is obtained via a [three-legged](#) OAuth flow.

Alternative Identifiers

Data Management IDs vs. Data Model IDs vs. ACC IDs

Object	Data Management API	Data Model APIs
Hub	b.561a419f-bd29-46bc-ab9f-4f4c27b8d33a	urn:adsk.ace:prod.scope:a1bca412-dcc2-42b5-8a4f-4bfcaeb8f6cc
Project	b.8c35aa99-e8af-457c-b397-ea94c0e44a42	urn:adsk.workspace:prod.project:cf013073-1a8d-48a3-b075-3b15fa27efa9
File Version = Element Group	urn:adsk.wipprod:fs.file:vf.85g5iiNeQCed28a2RzmsRw?version=2	YWVjZH5vaVM3ZG1J0XZRZ0dXV2ZMSllHM2pLX0wyQ344NWc1aWl0ZVFDZUQyOGEyUnptc1J3

Object	Data Management API	Autodesk Construction Cloud API
Project	b.8c35aa99-e8af-457c-b397-ea94c0e44a42	8c35aa99-e8af-457c-b397-ea94c0e44a42

Beta APIs

```
{  
  "errors": [  
    {  
      "message": "Field \"item\" argument \"hubId\" of type \"ID!\" is required, but it was not  
provided.",  
      "locations": [  
        {  
          "line": 2,  
          "column": 3  
        }  
      ],  
      "extensions": {  
        "code": "GRAPHQL_VALIDATION_FAILED"  
      }  
    },  
    {  
      "message": "Unknown argument \"projectId\" on field \"Query.item\".",  
      "locations": [  
        {  
          "line": 2,  
          "column": 8  
        }  
      ],  
      "extensions": {  
        "code": "GRAPHQL_VALIDATION_FAILED"  
      }  
    }  
  ]  
}
```

`item(
 projectId: ID!
 itemId: ID!
): Item` → `item(
 hubId: ID!
 itemId: ID!
): Item`



Beta APIs are subject to change. Please avoid using them in production environments.

APS support

- Email Support
- Documentation
- Tutorials
- Stack Overflow
- APS Accelerators
- APS @ AU

 [APS Support] Re: Support: Error when querying items (MFG V2 beta)

 Adam Nagy (Autodesk Platform Services - formerly Forge) <aps.help@autodesk.com>
To:  Christian Gessner

You replied to this message on Show Reply

##-- Please type your reply above this line --##

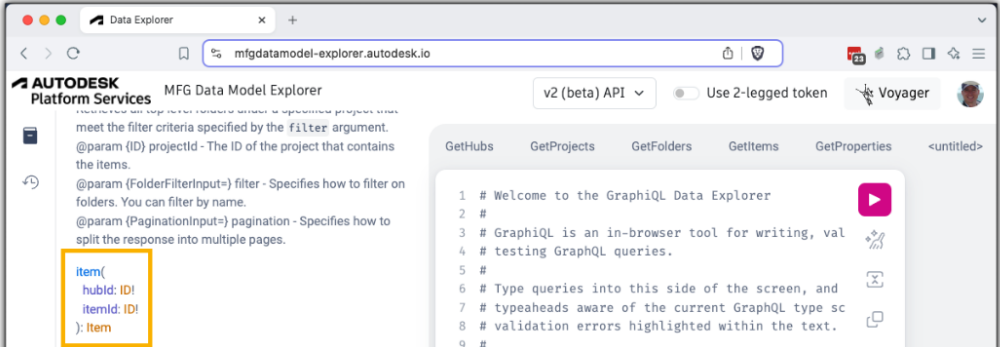
Your request (22279) has been updated. To add additional comments, reply to this email.

Reported: 3:56 PM
Answered: 4:06 PM

 Adam Nagy (APS Support)

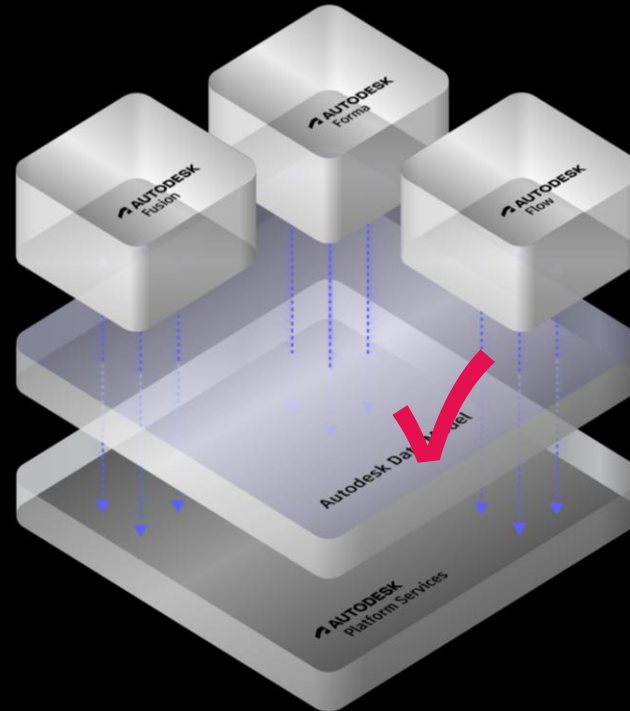
Hi Christian,

Yes, looks like a few query fields changed – you can always check the docs on <https://mfgdatamodel-explorer.autodesk.io/>



Autodesk Data Model - Conclusion

- ✓ Granular Data Queries
- ✓ Data Model Explorer
- ✓ Voyager
- ✓ Tutorials / Samples
- ✓ Support & Community



Resources on GitHub

- Manufacturing Data Model
 - Fusion – Business Central Integration
 - Fusion Add-In Source Code (Python)
 - Web Application Source Code (C#)
- AEC Data Model
 - ACC – Odoo Integration
 - Integration Source Code (PowerShell)
- Postman Collections



https://github.com/christiangessner/AU2024_SD2336

Join the Conversation!

<https://www.coolorange.com>

info@coolorange.com



COOL ORANGE

POWER UP YOUR DATA



<https://www.coolorange.com/au2024>

 AU 2024
THE DESIGN & MAKE CONFERENCE

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